

COSMOS

Cosmos bipinnatus / *Cosmos sulphureus*

Quick Facts

Scientific Name	Cosmos bipinnatus / Cosmos sulphureus
Family	Asteraceae (Daisy family)
Common Names	Cosmos, Mexican Aster, Garden Cosmos
Native Origin	Mexico, Central America, and northern South America
Colors	Pink, magenta, crimson, white, and bi-coloured (<i>Cosmos bipinnatus</i>); orange, yellow, and red (<i>Cosmos sulphureus</i>).
Height	60 to 150 cm. Dwarf varieties reach 30 to 40 cm.
Blooming Season	October to February in plains of India. In hill stations, June to October. Cosmos germinates and blooms readily within 8 to 10 weeks of sowing, making it one of the fastest garden flowers to bloom from seed.

Distribution in India

Commonly grown as a garden annual across India, particularly in the cooler months. Found in domestic gardens, public parks, and roadside plantings in West Bengal, Himachal Pradesh, Uttarakhand, Maharashtra, and Karnataka. Thrives at altitudes between 1,000 and 3,000 metres in hill stations like Darjeeling, Ooty, Kodaikanal, Shimla, and Mussoorie.

Description and Overview

Cosmos is a genus of flowering plants in the sunflower family Asteraceae, comprising about 40 species. The most commonly grown garden species are *Cosmos bipinnatus*, which produces pink, white, and crimson daisy-like flowers with yellow centres, and *Cosmos sulphureus*, which produces yellow, orange, and red flowers. The name 'cosmos' comes from the Greek word for harmony and order, which reflects the orderly arrangement of petals.

The plant has feathery, fern-like, bright green foliage that gives it an airy, delicate appearance. The flowers are held on long, slender stems above the foliage, making them excellent for cutting. Each flower head consists of 8 ray florets surrounding a cluster of disc florets at the centre.

Cosmos is one of the easiest annual flowers to grow from seed. It thrives in poor to moderately fertile, well-drained soil in full sun. Paradoxically, rich, overly fertilised soil produces lush foliage but fewer flowers. It self-seeds prolifically, meaning that once introduced to a garden, it often reappears year after year from dropped seeds.

In India, cosmos is particularly popular in cottage gardens and public parks during the winter season. It is a major pollinator plant, attracting butterflies, bees, and other insects. Hill stations such as Ooty

(Tamil Nadu), Kodaikanal, and Darjeeling are particularly known for mass cosmos plantings in their botanical gardens and parks during the blooming season.

Blooming Season and Conditions

October to February in plains of India. In hill stations, June to October. Cosmos germinates and blooms readily within 8 to 10 weeks of sowing, making it one of the fastest garden flowers to bloom from seed.

Cultivation

Seeds are sown directly in the garden bed or in trays between August and November for winter blooming in plains, or between March and May for summer blooming in hills. Seeds germinate in 5 to 10 days at 18 to 24 degrees Celsius. Thinning to 30 to 45 cm spacing is recommended to prevent overcrowding.

Cosmos requires minimal care once established. Deadheading (removing spent flowers) prolongs the blooming season significantly. The plant tolerates drought reasonably well but requires regular watering during establishment. It is largely pest-free, though aphids and spider mites can appear in dry conditions.

Uses and Applications

Ornamental: Used extensively in garden beds, borders, and as cut flowers in floral arrangements. **Pollinator gardens:** An important nectar source for butterflies, particularly the common tiger butterfly, and for various bee species. **Green manure:** The plant biomass is used as green manure and composted organic matter in home gardens. **Companion planting:** Grown alongside vegetables to attract pollinators and pest-controlling beneficial insects.

Interesting Facts

- Cosmos was introduced to Europe from Mexico in the 18th century and quickly became one of the most popular garden flowers worldwide.
- The plant is a symbol of peace, tranquillity, and love in many cultures, and is given as a gift to represent these feelings.
- A single cosmos plant can produce hundreds of flowers in a single season.
- *Cosmos sulphureus* contains yellow-orange carotenoid pigments that have been studied for potential use as natural food colouring agents.
- In Japan, there are entire fields of cosmos planted specifically as tourist attractions during the autumn season.

Source: Flower AI Knowledge Base — COSMOS entry. Scientific name: *Cosmos bipinnatus* / *Cosmos sulphureus*. Family: Asteraceae (Daisy family).